Three-dimensional MHD evolution of an ultramassive white dwarf with differential rotation and a mixed field

Initial and final states at 192^3 and 256^3 — interim report

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Abstract

A 2.005 $M_\odot$ equilibrium with strong differential rotation and a toroidal-dominated field, evolved in three-dimensional ideal MHD at two resolutions. **The star survives and keeps its mass; its ordered field does not.** At 192^3, run to $t = 60\,\mathrm{s}$, the magnetic energy falls by a factor of 2170 and the toroidal-to-poloidal amplitude ratio from 338 to 1.07, while the mass stays within 0.3%. The differential rotation is not erased — it *steepens*, $\Omega_{\rm out}/\Omega_{\rm core}$ falling from 0.34 to 0.27, because the transport takes angular momentum out of the outer layers and out of the star. The instability is $m = 1$ and grows in a few Alfvén times. Its rate is not converged — it rises 47% from 192^3 to 256^3 — and the post-saturation decay is not converged either, in the opposite direction. Both grids are now complete to $t = 12\,\mathrm{s}$. Averaged over the last two pulsation periods they agree on $\Omega_{\rm out}/\Omega_{\rm core}$ to 1.4% and on the peak field to 7%: the survival of the differential rotation is a result and not a resolution. What does not converge is the magnetic *energy*, twice as large on the finer grid at $t = 12\,\mathrm{s}$.

1 Initial state

1.1 The star

1.2 The fields

1.3 Equation of state and method

Mass	M	$2.0052\,M_\odot$
	$M/M_{\rm Ch}$ ($\mu_e = 2$)	**1.38**
Central density	ρ_c	3.00×10^9 g cm^{-3}
Mean density	$\langle\rho\rangle$	3.30×10^7 g cm^{-3}
Equatorial radius	$R_{\rm eq}$	3.917×10^8 cm
Polar radius	$R_{\rm pol}$	1.501×10^8 cm
Oblateness	$R_{\rm pol}/R_{\rm eq}$	**0.383**
Volume-equivalent radius	$R_{\rm vol}$	3.07×10^8 cm
Rotation law	$\Omega(\varpi) = \Omega_c A^2/(A^2 + \varpi^2)$	$A = 1.834 \times 10^8$ cm
Central angular velocity	Ω_c	8.193 rad s^{-1} ($P = 0.77$ s)
Equatorial angular velocity	$\Omega(R_{\rm eq})$	1.47 rad s^{-1} ($P = 4.3$ s)
Degree of differential rotation	$\Omega_c/\Omega(R_{\rm eq})$	**5.6**
Equatorial velocity	$v_{\rm eq}$	6.2×10^3 km s^{-1} $= 0.021\,c$
Rotational support	$T/\lvert W\rvert$	0.0993
Mass shedding	$\Omega_{\rm eq}^2 R_{\rm eq}/g_{\rm eq}$	0.469
Dynamical time	$t_{\rm dyn}$	0.475 s
Alfvén time	t_A	2.383 s
Central rotation period	$P_{\rm rot}$	0.767 s

Table 1: The configuration. It lies 38% above the Chandrasekhar mass and can only exist because of the differential rotation, which is also why it is so flattened. It is not at breakup (47% of shedding) and is below both bar-mode thresholds ($T/|W| = 0.099$ against 0.14 secular and 0.27 dynamical). The surface period of 4.3 s is some three times shorter than the fastest white dwarf spin claimed in the literature; this is a merger-remnant-like configuration, not a model of an observed star.

	analytic model	192^3 grid	256^3 grid
$\max\lvert B_\varphi\rvert$ (G)	3.231×10^{13}	3.196×10^{13}	3.204×10^{13}
$\max\lvert B_{\rm pol}\rvert$ (G)	7.63×10^9	9.46×10^{10}	8.41×10^{10}
$\max\lvert B\rvert/B_c$	0.732	0.724	0.726
$E_{\rm mag}$ (erg)	6.035×10^{49}	6.035×10^{49}	6.056×10^{49}
$E_{\rm tor}/E_{\rm mag}$	0.99999995	0.999997	0.999998
$E_{\rm tor}/E_{\rm pol}$	$\mathbf{2.18 \times 10^7}$	3.11×10^5	5.22×10^5
$\max\lvert B_\varphi\rvert/\max\lvert B_{\rm pol}\rvert$	**4234**	338	381
$E_{\rm pol}/\lvert W\rvert$	5.2×10^{-10}	—	—
Toroidal filling factor	—	1.23%	—

Table 2: Mixed field, toroidal-dominated to one part in 10^7 by energy: a 10^9 G exterior dipole beside a 3×10^{13} G interior toroidal field. The peak toroidal field is reconstructed on the grid to 98.9% of the model value at 192^3; the poloidal component is not, because interpolating an axisymmetric model onto a Cartesian mesh injects poloidal field an order of magnitude above the true one. *The poloidal quantities at $t = 0$ are therefore numerical floors, not physics*, and the growth measured later is counted against them. The filling factor is $8\pi E_{\rm tor}/\max|B_\varphi|^2$ over the stellar volume.

Equation of state	`ztwd`: zero-temperature degenerate electrons, $\mu_e = 2$
	barotropic, $P = P(\rho)$; all $\partial/\partial T$ vanish, $c_V = c_P = 0$
Code	Castro 26.07 / AMReX; ideal MHD, constrained transport, HLLD
Grid	3D Cartesian, single level, box 1.8×10^9 cm per side
	192^3: $\mathrm{d}x = 9.375 \times 10^6$ cm 256^3: 7.031×10^6 cm
Gravity	`PoissonGrav`, full 3D solve (the quadrupole is large at $R_{\rm pol}/R_{\rm eq} = 0.38$)
Initial condition	Grad–Shafranov/SCF on 385×769 (ϖ, z), interpolated from $\mathbf{A}$
CFL	0.5 at 192^3; 0.3 at 256^3 past $t \simeq 4.8$ s
Floor / sponge	$\rho_{\rm min} = 10^2$ against ambient 2×10^4; sponge 10^4–10^5
Diagnostics	rebuilt from B_x, B_y, B_z; the native `emag_density`,
	`etor_density` and `Div_B` derives are corrupt (see Sec. 5)

2 Final states

2.1 Same instant, both grids

$t \simeq 5.2\,\mathrm{s}$	192^3	256^3	ratio
E_{tor} (erg)	1.965×10^{49}	1.784×10^{49}	0.91
E_{pol} (erg)	1.791×10^{48}	2.510×10^{48}	1.40
$E_{\mathrm{mag}}/E_{\mathrm{mag}}(0)$	35.5%	33.6%	
$E_{\mathrm{tor}}/E_{\mathrm{pol}}$	11.0	7.1	
$\max\lvert B_\varphi\rvert$ (G)	4.539×10^{13}	4.045×10^{13}	0.89
$\max\lvert B_{\mathrm{pol}}\rvert$ (G)	3.827×10^{13}	3.104×10^{13}	0.81
$\max\lvert B_\varphi\rvert/\max\lvert B_{\mathrm{pol}}\rvert$	1.19	1.30	
$\max\lvert B\rvert/B_c$	1.118	0.931	
M ($M_\odot$)	2.0035	2.0031	
R_{vol} (10^8 cm)	3.51	2.97	

Table 3: The two grids at the same physical time, at the end of the growth phase. They agree on the state to within tens of percent, but not on the rate that got there — see Sec. 4.

2.2 The common window, $t = 9$–$12\,\mathrm{s}$

The 256^3 run has since reached its 12 s target, so the two grids can be compared over a window rather than at an instant. That matters: the star breathes by ±14% every 1.498 s, and any single snapshot catches an arbitrary phase of it. Table 4 averages over the last two periods, which removes the pulsation and leaves the state.

mean over 9–12 s	192^3	256^3	256/192
The star — converged			
M ($M_\odot$)	2.0028	2.0042	1.0007
R_{vol} (10^8 cm)	3.113	3.049	0.98
V/V_0	1.06	0.99	0.93
Ω_{core} ($\mathrm{rad\,s^{-1}}$)	6.654	6.733	**1.012**
Ω_{out} ($\mathrm{rad\,s^{-1}}$)	2.294	2.357	**1.027**
$\Omega_{\mathrm{out}}/\Omega_{\mathrm{core}}$	0.345	0.350	**1.014**
L_z (10^{50} g cm^2 s^{-1})	2.168	2.078	0.96
$\max\lvert B\rvert/B_c$	0.771	0.826	1.07
The field — not converged			
E_{tor} (erg)	4.07×10^{48}	8.00×10^{48}	**1.96**
E_{pol} (erg)	3.07×10^{47}	8.24×10^{47}	**2.69**
$E_{\mathrm{mag}}/E_{\mathrm{mag}}(0)$	7.3%	14.6%	2.01
$E_{\mathrm{tor}}/E_{\mathrm{pol}}$	13.3	9.7	0.73
$L_z^{\mathrm{outer}}/L_z^{\mathrm{inner}}$	5.04	5.90	1.17

Table 4: The two grids over the same three seconds, averaged over two pulsation periods. The split is sharp and it is the main new result of this report: everything about *the star* agrees to a few percent — mass, size, and above all the rotation, where the core, the outer shell and their ratio all land within 3% — while everything about the *energy* in the field differs by a factor of two. The finer grid dissipates less, which is exactly what an ideal-MHD run with no explicit resistivity should do.

The rotation rows are the ones to take away. The conclusion that the star keeps its differential rotation — the property that lets it hold $2\,M_\odot$ — was previously measured at one resolution only. It is now measured at two, and they agree to 1.4%.

2.3 The end state, 192^3

Only the 192^3 run has reached a steady late state; it was carried to $t = 60\,\mathrm{s}$ and stopped changing well before that.

192^3	$t = 0$	$t = 38.5\,\mathrm{s}$	change
The star			
M ($M_\odot$)	2.00667	2.00618	**−0.02%**
R_{vol} (10^8 cm)	3.068	2.716	−11%
R_{eq} (10^8 cm)	3.750	3.204	−15%
$R_{\mathrm{pol}}/R_{\mathrm{eq}}$	0.425	0.556	rounder
Ω_{core} ($\mathrm{rad\,s^{-1}}$)†	7.86	7.59	−3%
Ω_{out} ($\mathrm{rad\,s^{-1}}$)†	2.56	1.92	−25%
$\Omega_{\mathrm{out}}/\Omega_{\mathrm{core}}$†	0.341	0.266	**−22%**
L_z (10^{50} g cm^2 s^{-1})†	1.979	2.101	−5.5% from peak
$L_z^{\mathrm{outer}}/L_z^{\mathrm{inner}\dagger}$	6.35	4.35	**−32%**
I (10^{49} g cm^2)†	4.16	4.59	−8.8% from $t = 12$
The field			
E_{mag} (erg)	6.035×10^{49}	2.782×10^{46}	/ **2170**
E_{tor} (erg)	6.035×10^{49}	2.658×10^{46}	/ 2270
E_{pol} (erg)	1.942×10^{44}	1.236×10^{45}	×6.4
$E_{\mathrm{tor}}/E_{\mathrm{pol}}$	3.11×10^5	21.5	/ 1.4×10^4
$\max\lvert B_\varphi\rvert$ (G)	3.196×10^{13}	1.227×10^{12}	/ 26
$\max\lvert B_{\mathrm{pol}}\rvert$ (G)	9.46×10^{10}	1.148×10^{12}	×12
$\max\lvert B_\varphi\rvert / \max\lvert B_{\mathrm{pol}}\rvert$	**338**	**1.07**	/ 316
$\max\lvert B\rvert/B_c$	0.724	0.031	

Table 5: Initial and end state at 192^3. The star keeps its mass to 0.02% and *strengthens* its differential rotation; the field is reduced by three orders of magnitude in energy and ends with its two components equal in peak strength. †Rotation and L_z are quoted at $t = 60\,\mathrm{s}$, the end of the run; the field and shape columns at $t = 38.5\,\mathrm{s}$, the last instant processed with the field diagnostics.

Three points in Table 5 carry most of the physics.

The mass does not move. 0.02% from start to end, 0.3% in amplitude along the way, while the magnetic energy of the same star falls by a factor of 2170. Whatever destroys the field does not unbind the star.

Losing L_z and spinning up is expected here. $L_z = I\bar{\Omega}$ and I is not constant: between $t = 12$ and $60\,\mathrm{s}$ the star loses 2.8% of its angular momentum, contracts enough to lose 8.8% of its moment of inertia, and gains 6.6% in $\bar{\Omega}$ (Fig. 1b). This is not an oddity of the simulation. It is the behaviour known for rotating white dwarfs near the maximum mass, where a configuration shedding angular momentum contracts and spins up rather than spinning down — the spin-up branch discussed by Boshkayev et al. (2013). Their analysis is uniformly rotating and general relativistic while this configuration is differentially rotating, Newtonian and magnetised, so the correspondence is in the mechanism and not in the numbers.

The differential rotation not only survives, it strengthens. Fig. 1 is the diagnostic the long run was extended to provide. The magnetic transport is real and its signature is unambiguous: L_z falls at $0.21\%\,\mathrm{s}^{-1}$ while the toroidal field lives and $0.10\%\,\mathrm{s}^{-1}$ after it is gone. With no explicit viscosity and no explicit resistivity in these runs, nothing else in the scheme brakes anything, so that factor of two identifies the agent.

But the transport does not homogenise the star. Had it done so, $\Omega_{\mathrm{out}}/\Omega_{\mathrm{core}}$ would climb towards unity; it falls instead, from 0.34 to 0.27. The core holds its angular velocity to 3% over $60\,\mathrm{s}$ while the outer shell loses 25% of its own — angular momentum is carried outward and out of the star, and it is the outer layers that pay. Split by mass rather than by radius the same thing shows: $L_z^{\rm outer}/L_z^{\rm inner}$ falls from 6.35 to 4.35, because the outer half of the mass loses 1% of its angular momentum while the star as a whole loses 5.5% and the inner half gains. *The star survives because the field never came close to removing what supports it, and in the time it existed it steepened the profile rather than flattening it.*

"Destroyed" means disordered, not weakened — until late. At $t = 5.2\,$s the peak toroidal field is *higher* than initially, 4.54×10^{13} against $3.20 \times 10^{13}\,$G, while the energy has fallen by a factor of three: the field is concentrating into filaments, its effective volume dropping from 1.49×10^{24} to $2.40 \times 10^{23}\,\mathrm{cm}^3$. Only afterwards does the peak collapse as well, to $1.23 \times 10^{12}\,$G.

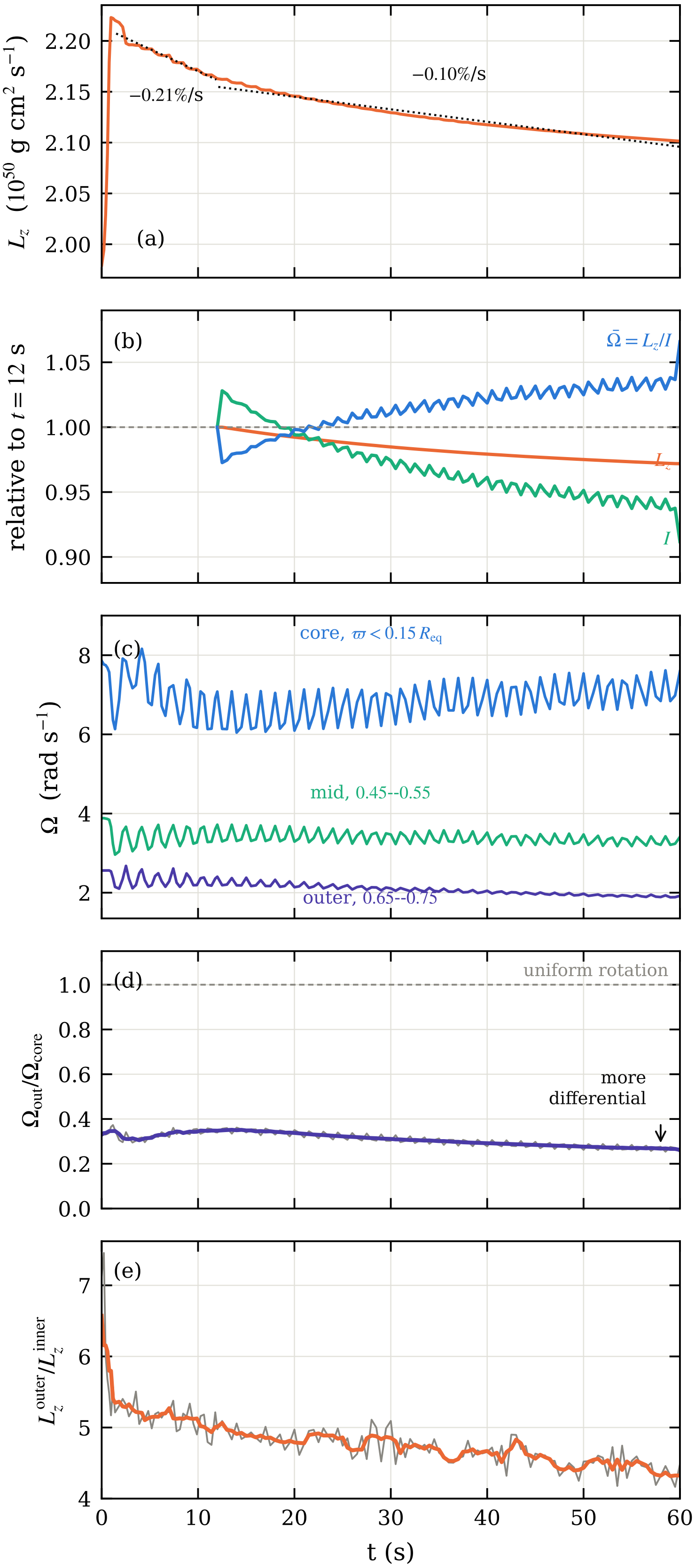

Figure 1: 192^3 over the full 60 s. **(a)** L_z of the star, with the two braking rates fitted; the 12% rise in the first second is the initial transient settling. **(b)** the decomposition $L_z = I\bar{\Omega}$, normalised at $t = 12$ s, which answers the obvious objection to (a): the star loses angular momentum and its angular velocity rises anyway, because I is not constant. Between $t = 12$ and 60 s, L_z falls 2.8%, I falls 8.8% as the star contracts, and $\bar{\Omega} = L_z/I$ gains 6.6% — a skater, minus the losses. **(c)** Ω mass-weighted in three cylindrical shells, all chosen to stay inside the star at every phase of the 1.5 s pulsation — note they sit at fixed fractions of the *initial* R_{eq}, so part of the decline in the outer one is the star having shrunk out from under it. **(d)** the differential rotation; uniform rotation would be 1 and the star moves away from it. **(e)** L_z of the outer half of the *mass* over that of the inner half —6 split by mass rather than radius, with the dividing radius recomputed at every snapshot, because a shell at fixed ϖ would measure the pulsation instead of the transport. In (c) and (d) a running mean over one pulsation period is drawn over the raw curve; the ripple is the pulsation and aliases badly against coarse sampling.

3 The evolution between them

The field goes through three phases, all visible in Fig. 2.

Phase	Window	Rate	Signature
Growth	$t = 1.3$–$4.0\,\mathrm{s}$	$\sigma = 1.369\ \mathrm{s}^{-1}$ $(r = 0.994)$ e-fold of amplitude $0.61\,t_A$	E_{pol} exponential; $m = 1$ takes over
Saturation	$t \simeq 5.3\,\mathrm{s}$	—	E_{pol} peaks at $10^4\times$ its floor; $\max B_\varphi / \max B_{\mathrm{pol}} \to 1.19$
Decay	$t = 5.4$–$30\,\mathrm{s}$	e-fold 3.6–4.0 s	E_{tor} falls 10^3
Residual	$t > 30\,\mathrm{s}$	e-fold 37 s	$E_{\mathrm{mag}} \simeq 3 \times 10^{46}$ erg, flat

Table 6: Phases of the field evolution at 192^3. The decay stalls after $t \simeq 30\,\mathrm{s}$: the field does not go to zero but settles at a disordered residual of order 10^{12} G.

The mode is $m = 1$. An azimuthal Fourier projection gives $|m{=}1|/|m{=}0| = 1.2 \times 10^{-16}$ at $t = 0$ — the initial condition is axisymmetric to machine precision. From $t \simeq 2.5\,\mathrm{s}$ the poloidal field is $m = 1$ dominated, with $|B_{z,m=1}|/|B_{z,m=0}|$ reaching 21 at $t = 4\,\mathrm{s}$, and the toroidal $m = 1$ saturates at $\simeq 10\%$ of the axisymmetric component. This rules out the axisymmetric MRI, which could not exist here in any case: built from the *vertical* field, $\lambda_{\mathrm{MRI}} = 2.9 \times 10^4\,\mathrm{cm}$, or 0.003 of a cell. What remains is the Tayler kink or the non-axisymmetric toroidal-field MRI, which the mode number alone cannot separate.

The star pulsates and does not stop. Past the initial transient, $\rho_{\max}$ settles into a clean periodic oscillation of period 1.498 s ($3.15\,t_{\mathrm{dyn}}$, 1.95 rotations) at $\pm 14.4\%$, with an envelope e-folding near 140 s. It is still running at $t = 60\,\mathrm{s}$.

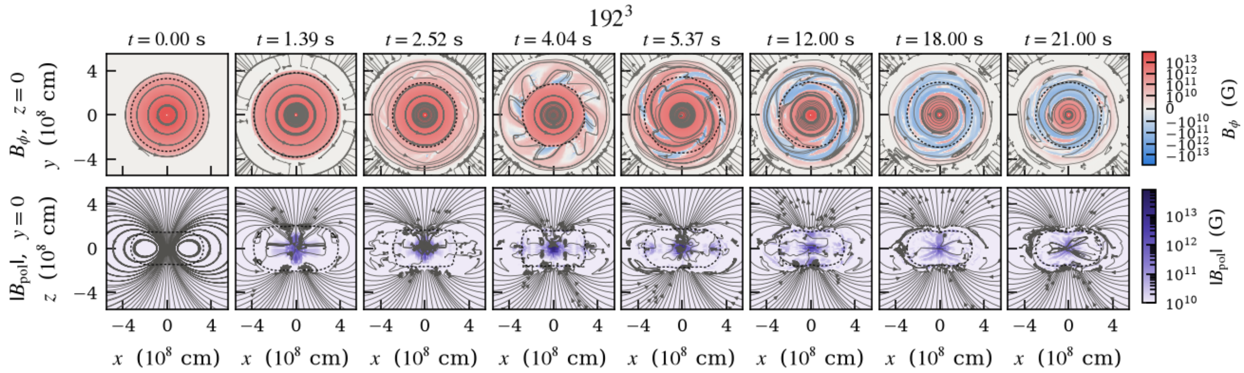

Figure 2: 192^3. **Top:** equatorial cut $z = 0$, where B_φ lies in the plane, so the streamlines *are* the toroidal field lines — circles while the star is axisymmetric. **Bottom:** meridional cut $y = 0$, where the in-plane field *is* the poloidal one. Colour is B_φ and $|B_{\rm pol}|$ respectively, both starting at 10^{10} G so the two rows are comparable; the near-empty poloidal panel at $t = 0$ is the statement that the poloidal field is three decades below the toroidal one. Same scales as Fig. 5.

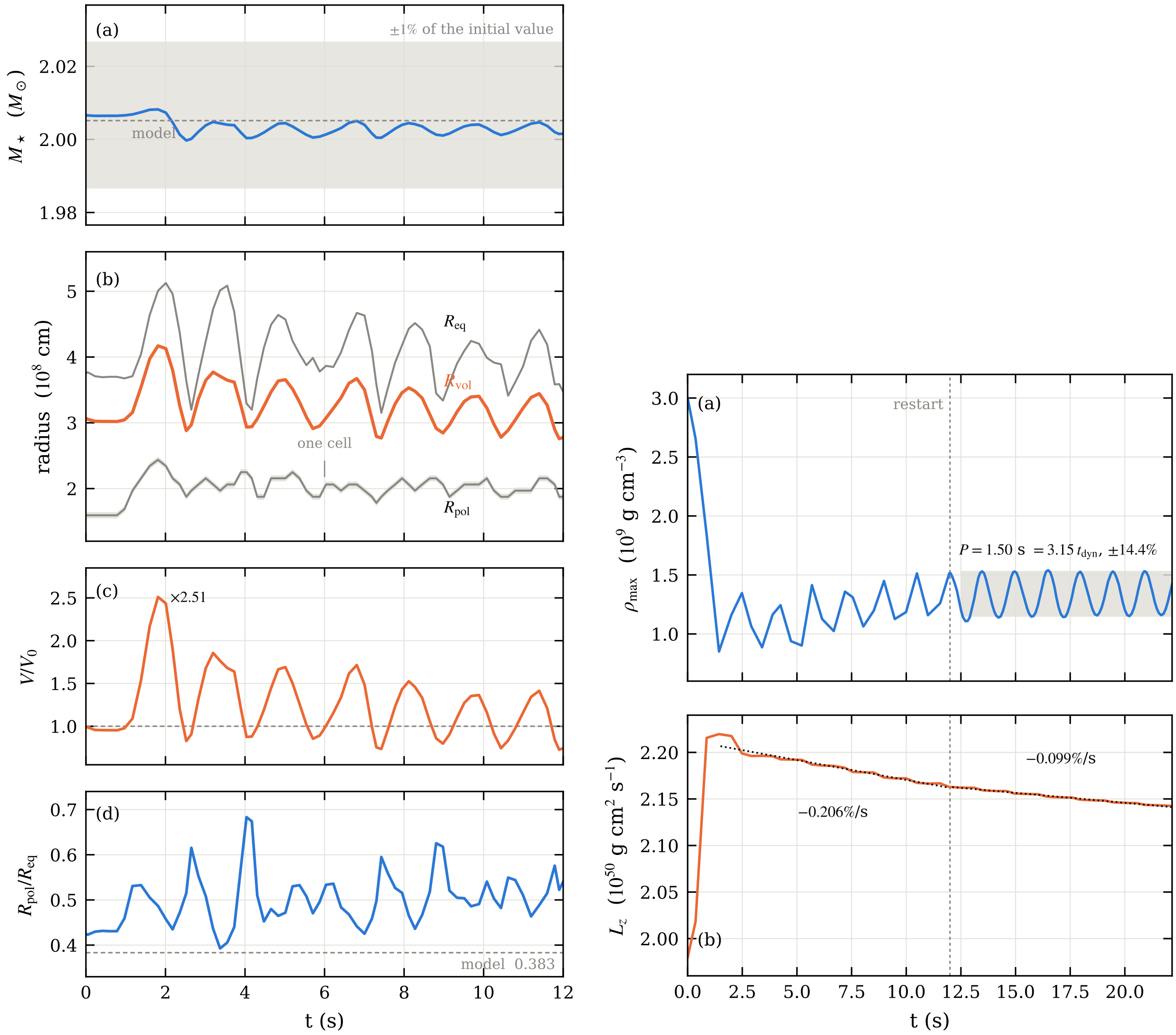

Figure 3: Left, 192^3 **to** $t = 12\,$**s:** mass against a $\pm 1\%$ band; the three radii with one grid cell drawn around $R_{\rm pol}$; volume; oblateness. **Right, to** $t = 22\,$**s:** central density, and axial angular momentum with the two braking rates fitted. The dashed vertical is the restart at $t = 12\,$s.

4 Numerical convergence

	192^3	256^3	verdict
Mass range ($M_\odot$)	1.9997–2.0082	2.0021–2.0073	converged
$R_{\rm vol}(t=0)$ (10^8 cm)	3.068	3.069	converged (0.03%)
$\Omega_{\rm core}$, 9–12 s (rad s^{-1})	6.654	6.733	converged (1.2%)
$\Omega_{\rm out}/\Omega_{\rm core}$, 9–12 s	0.345	0.350	**converged** (1.4%)
$\max\|B\|$, 9–12 s (G)	3.40×10^{13}	3.65×10^{13}	converged (7%)
σ from $E_{\rm pol}$ (s^{-1})	1.369	2.007	+47%**, not converged**
σ from the $m=1$ mode (s^{-1})	1.968	2.752	+40%, same factor
$E_{\rm pol}(t=0)$, the floor (erg)	1.94×10^{44}	1.16×10^{44}	finer seed is smaller
$\min(\max B_\varphi / \max B_{\rm pol})$	1.19	**0.805**	instability goes further
Decay rate γ of $E_{\rm tor}$, 6–12 s	$0.282\,{\rm s}^{-1}$	$0.141\,{\rm s}^{-1}$	/ 2.0**, not converged**
as an e-folding	3.5 s	7.1 s	
$E_{\rm mag}$ at $t = 12$ s (erg)	4.4×10^{48}	8.8×10^{48}	×2.0
First volume peak	$2.51\,V_0$	$1.79\,V_0$	early pulsation is numerical

Table 7: Three verdicts. *Converged:* everything about the star — mass, size, peak field, and the whole rotation profile. *Not converged:* the growth rate rises with resolution while the decay rate falls, each by about a factor of two per refinement, so the measured growth is a lower bound and the measured decay an upper bound; the coarse grid exaggerates the violence at both ends. The early large-amplitude pulsation is largely the initial condition ringing on the mesh.

There is no explicit resistivity in these runs, so the decay of $E_{\rm tor}$ *is* the numerical resistivity, and the two grids measure it. Halving γ while dx falls by a factor 1.33 puts it at d$x^{2.4}$, consistent with the second-order scheme and extrapolating to zero. Read literally, that says the field decay seen here is a resolution artefact and the physical star would keep its field far longer — but a two-point fit cannot distinguish that from a decay that converges to something small but finite, and the growth rate moving the other way (d$x^{-1.3}$) warns that both ends of the evolution are still mesh-limited.

Settling the rate would need 384^3, about five times the cost of 256^3. The alternative, and probably the right one, is explicit resistivity, giving the problem a physical dissipation scale instead of letting the mesh set it; Castro does not currently implement it.

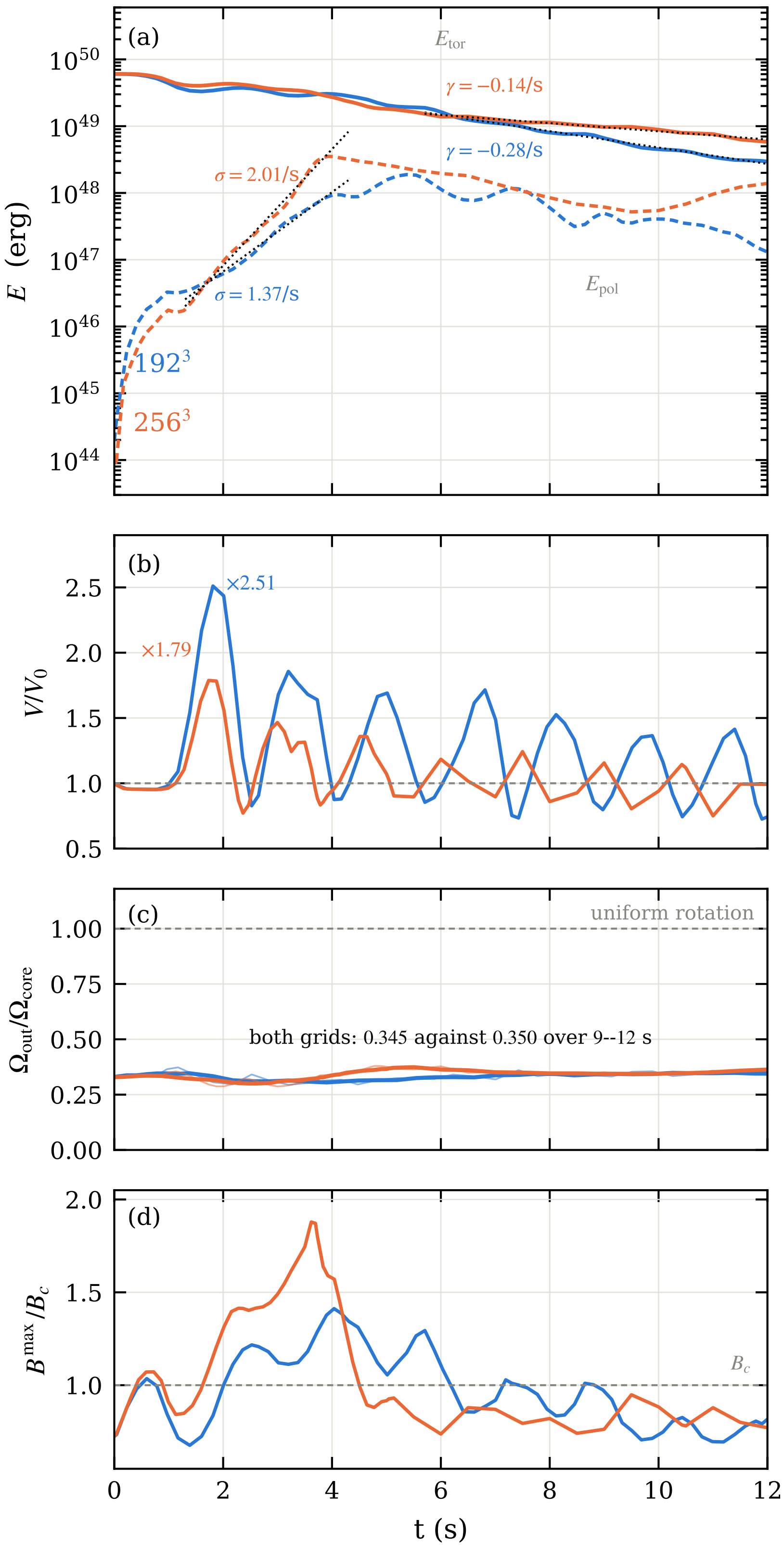

Figure 4: 192^3 against 256^3 over the full 12 s. **(a)** Toroidal (solid) and poloidal (dashed) energy, with the fitted growth σ and decay γ. The two rates move in *opposite* directions with refinement. **(b)** V/V_0: the first peak drops from 2.51 to 1.79, its excess over unity scaling as $\mathrm{d}x^{2.3}$. **(c)** The differential rotation — neither grid approaches uniform rotation and the two agree to 1.4%. **(d)** Peak field against B_c; above the Landau field for roughly half the run at both resolutions.

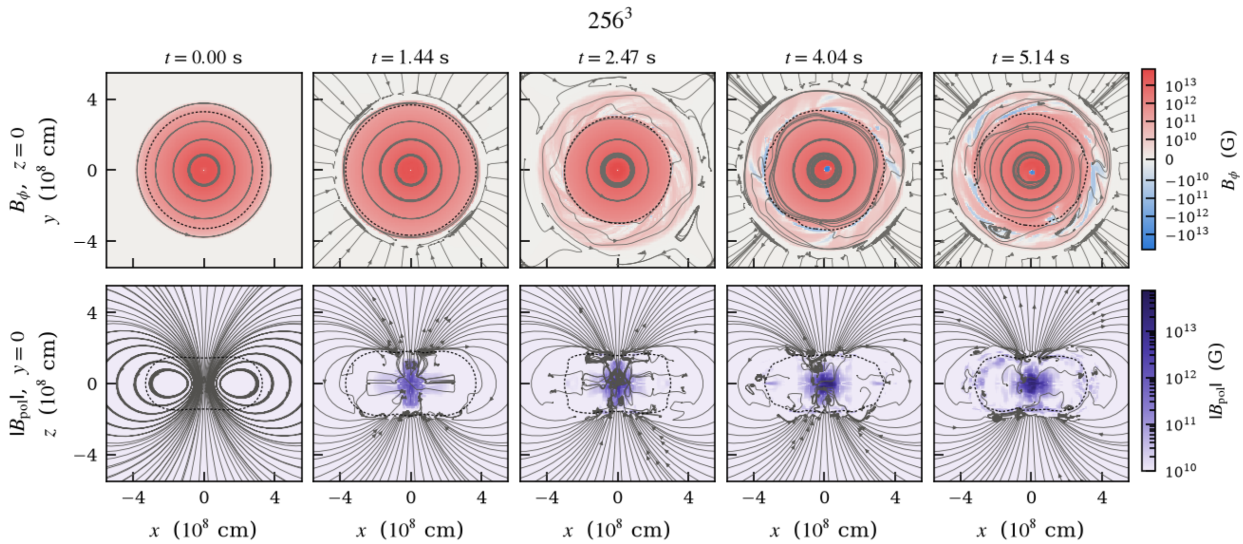

Figure 5: 256^3, same layout, planes and colour scales as Fig. 2.

5 Caveats

1. **The EOS is barotropic.** `ztwd` has no thermal reservoir, so the 6×10^{49} erg that leave the field have no physical destination and are dissipated at the grid scale. We can say the ordered field is destroyed; we cannot say where the energy went. The stratification is also neutral, with no buoyancy to limit the mode — in a stratified star the rate would be lower, making σ a *physical* upper bound while it is a *numerical* lower bound.
2. **Part of the run is outside the EOS validity range.** $\max|B|$ reaches $1.41\,B_c$ at $t = 4.0$ s at 192^3 and $1.88\,B_c$ at $t = 3.6$ s at 256^3, exceeding the Landau critical field in 44% and 48% of the sampled instants respectively. This does *not* improve with resolution — the finer grid goes further above B_c, not less far. The late state, at $0.03\,B_c$, is well inside.
3. **The poloidal field at $t = 0$ is a numerical floor.** Cartesian interpolation of an axisymmetric model injects $E_{\rm pol}$ some 70 times above the model's value even at 256^3. All poloidal growth is measured against that floor, which is why it is quoted alongside.
4. **Only 192^3 goes past 12 s.** The two grids now overlap fully to $t = 12$ s, but the late evolution — the settled 1.50 s pulsation, the halving of the braking rate once the field is gone, the steepening of the differential rotation to $\Omega_{\rm out}/\Omega_{\rm core} = 0.27$, and the $L_z = I\bar{\Omega}$ spin-up — is measured at 192^3 alone. Within the common window the two grids agree on the rotation to 1.4%, which is reassuring but is not a check on what happens after it.
5. **Native magnetic diagnostics are unusable.** `emag_density`, `etor_density` and `Div_B` pack face-centred components of different index types into one cell-centred FAB and read past the end of the box, `Div_B` reaching 3×10^{146} where it should be ~ 0. Everything here is rebuilt from B_x, B_y, B_z, which are correct.

Status

192^3: complete, $t = 60$ s. 256^3: complete, $t = 12$ s (7059 steps).

Inputs, submission scripts, diagnostic tools, data and figure code are under version control in the `WD_MAG` repository.

References

Boshkayev, K., Rueda, J. A., Ruffini, R., & Siutsou, I. 2013, *On General Relativistic Uniformly Rotating White Dwarfs*, ApJ, 762, 117.